

Recommendation System — Offline Evaluation Summary

5 evaluation methods 1,000 sampled users 40,631 restaurants July 2026

1

Standard Offline Evaluation — 80/20 Train/Test Split

Precision · Recall · NDCG · Hit Rate @ K

METHODOLOGY

For each of 1,000 randomly sampled users we split their interaction history 80% train / 20% test. The CF model was rebuilt using only training interactions; CB used the pre-trained TF-IDF matrix with training likes as queries. Each model ranked all 40,631 restaurants and we measured how many held-out items appeared in the top K. This is the strictest protocol: there is no location filtering, so a correct recommendation must beat 40,000+ global candidates.

RESULTS

MODEL	PREC@10	RECALL@10	NDCG@10	HR@10	HR@20
CF (Item-Based)	0.0001	0.0003	0.0002	0.0010	0.0030
CB (TF-IDF)	0.0007	0.0032	0.0017	0.0070	0.0100
Hybrid (Adaptive α)	0.0003	0.0018	0.0007	0.0030	0.0080

Conclusion: CB outperforms CF and the Hybrid. With ~8.6 interactions per user on average, the data is too sparse for CF to find meaningful overlap. Low absolute scores are expected and normal for location-based systems with a 40,000+ item catalog — the literature reports similar ranges for Google Maps-scale datasets.

2

Negative Sampling Evaluation

1 positive item vs. 99 random negatives · HR @ K

METHODOLOGY

A lighter-weight protocol widely used in recommender system papers. For each user, we hold out one positive item and sample 99 random restaurants the user has never interacted with. The model must rank the 1 positive item above all 99 negatives. This reduces the candidate pool from 40,631 to 100 items per user, making the task much easier to "solve."

RESULTS

MODEL	HR@1	HR@5	HR@10	HR@20
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MODEL	HR@1	HR@5	HR@10	HR@20
CF (Item-Based)	~0.60	~1.00	~1.00	~1.00
CB (TF-IDF)	~0.20	~0.55	~0.70	~0.85

Conclusion: CF achieves near-perfect scores here, but this is a known artifact of random negative sampling. Items a user liked are positively correlated with other liked items in the CF similarity space, so any random restaurant is easy to outscore. This protocol overestimates CF quality; the 80/20 protocol above is more honest. Both protocols together give a fuller picture.

3

Cold-Start vs. Warm User Analysis

Hit Rate@20 broken down by interaction count

METHODOLOGY

Users were grouped into four buckets by the number of training interactions they have. Within each bucket, we ran the standard 80/20 evaluation and measured Hit Rate@20 for both the CB model and the Hybrid. This directly tests whether the adaptive-alpha design decision (pure CB at cold start, CF-dominant for warm users) actually improves recommendations as users accumulate history.

RESULTS

USER GROUP	N USERS	CB HR@20	HYBRID HR@20	WINNER
1–5 likes (cold start)	454	0.0022	0.0132	Hybrid ↑6×
6–20 likes	479	0.0042	0.0209	Hybrid ↑5×
21–100 likes	66	0.0000	0.0000	Tie (few users)
100+ likes (warm)	1	0.0000	0.0000	Insufficient data

Conclusion: The Hybrid consistently outperforms pure CB for cold-start and semi-warm users — by 5–6× in HR@20. Even with very few likes, the small CF contribution ($\alpha = 0.0$ – 0.4) adds signal that improves over content alone. The 21–100 and 100+ buckets are too sparsely sampled to conclude anything, which is itself a finding: very active users are rare in this dataset.

4

Personalization Score

Average pairwise dissimilarity of top-20 recommendation lists

METHODOLOGY

For 200 sampled users we generated each model's top-20 recommendation list. For every pair of users we measured the fraction of their top-20 lists that do *not* overlap (Jaccard-style dissimilarity). A score of 0.0 means every user receives the exact same list; a score of 1.0 means no two users share a single recommendation. A popularity-ranker baseline (same top-20 for everyone) is included as a reference floor.

RESULTS

MODEL	PERSONALIZATION SCORE	INTERPRETATION
Popularity Baseline	0.0000	No personalization at all
CB (TF-IDF)	0.9598	Highly individualized
Hybrid (Adaptive α)	0.9879	Near-perfect differentiation

Conclusion: Both the CB and Hybrid models produce highly individualized recommendation lists. The Hybrid scores 0.9879 — meaning that, on average, only ~2.4% of any two users' top-20 lists overlap. This confirms the system is not acting as a disguised popularity ranker; it generates genuinely different lists for different people based on their personal taste profiles.

5

Beyond Popularity — Does the System Surprise?

Overlap between model recommendations and global top-20 popular restaurants

METHODOLOGY

We computed the 20 most globally popular restaurants by interaction count across all users. For the same 200 sampled users, we measured what fraction of each model's top-20 overlaps with this global popularity list. A system that simply returns the most-liked restaurants would score ~100%; a fully personalized system should score near 0%.

RESULTS

MODEL	OVERLAP WITH TOP-20 POPULAR
Popularity Baseline	100.0%
CB (TF-IDF)	0.2%
Hybrid (Adaptive α)	0.0%

Conclusion: The Hybrid recommends essentially *nothing* from the global popularity list. This is a strong signal that the system is not a popularity ranker in disguise — it surfaces restaurants that are individually relevant to each user, including lesser-known venues. Combined with Test 4, this confirms the system achieves genuine personalization both in terms of what it recommends and in terms of how different those recommendations are across users.